

Careless/Insufficient Effort Responding (C/IER) Detection

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This notebook presents a tutorial on the detection of careless or insufficient effort responding (C/IER) in self-report questionnaires. We use real database from WHOQOL-Bref scale as an example, and we apply the ‘careless’ package and Laz.R (Biemann et al., 2025) to detect C/IER (post hoc indicators). We also discuss the importance of procedural C/IER detection in the quality control of data obtained through self-report questionnaires.

1 Contextualization

The detection of Careless or Insufficient Effort Responding (C/IER) is an essential step in the quality control of data obtained through self-report questionnaires. C/IER occurs when respondents complete the instrument without attention to the item content, compromising the validity of the psychometric and substantive inferences derived from the data (Curran, 2016; Huang et al., 2026).

The analytical flow of this document follows three sequential levels of detection:

1. **Questionnaire:** procedural exclusions (a priori), including dropout, response time, and quality control items.
2. **WHOQOL Scale:** diagnostic post-hoc indicators (Laz.R, Longstring, Mahalanobis Distance, and IRV) with cut-off points estimated via the Kneedle algorithm (Biemann et al., 2025).
3. **Items:** VIF-based multicollinearity diagnosis to anticipate convergence problems in CFA.

2 Data reading and preparation

The `data.csv` file contains the raw survey responses (separator `;`). In this step, dropout records are removed, variables for the analytical flow are selected, and the total response time (`TIME`) is derived.

Rows: 1,326

Columns: 33

```
$ ID      <dbl> 10441016017, 10440981694, 10440601239, 10440562981, 104403585...
$ CREATED <chr> "28/12/2018 12:00", "28/12/2018 11:25", "28/12/2018 03:23", "...
$ MODIFIED <chr> "28/12/2018 12:20", "28/12/2018 11:47", "28/12/2018 03:39", "...
$ CQ1     <dbl> 1, 1, 1, 1, 1, 1, 1, 1, NA, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...
$ CQ2     <dbl> 1, 1, 1, 1, 1, 1, 1, 1, NA, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...
$ CQ3     <dbl> 1, 1, 1, 1, 1, 1, 1, 1, NA, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...
$ Q1      <dbl> 5, 3, 1, 4, 3, 4, 3, 4, 2, 5, 4, 3, 4, 2, 4, 5, 4, 5, 4, 4, 4...
$ Q2      <dbl> 5, 4, 1, 4, 4, 4, 4, 4, 2, 5, 4, NA, 4, 1, 4, 5, 4, 4, 4, 4, ...
$ Q3      <dbl> 1, 2, 3, 2, 1, 1, 1, 1, 4, 2, 3, 2, 2, 3, 3, 1, 2, 2, 2, 3, 1...
$ Q4      <dbl> 1, 2, 2, 1, 1, 1, 2, 1, 4, 1, 3, 2, 2, 2, 2, 2, 2, 1, 2, 1, 1...
$ Q5      <dbl> 3, 3, 1, 3, 3, 3, 3, 3, 1, 4, 3, 4, 4, 1, 3, 4, 3, 5, 3, 4, 3...
$ Q6      <dbl> 4, 4, 1, 3, 5, 4, 3, 4, 4, 4, 5, 3, 3, 4, 4, 5, 3, 5, 4, 4, 3...
$ Q7      <dbl> 4, 3, 1, 3, 5, 4, 2, 3, 2, 4, 3, 2, 4, 2, 3, 4, 3, 4, 2, 3, 4...
$ Q8      <dbl> 1, 2, 1, 2, 2, 4, 4, 4, 2, 3, 2, 3, 3, 3, 2, 2, 4, 3, 3, 3, 2...
$ Q9      <dbl> 5, 4, 1, 2, 3, 4, 3, 4, 3, 3, 3, 3, 4, 1, 2, 3, 4, 4, 2, 3, 2...
$ Q10     <dbl> 5, 3, 1, 3, 5, 4, 2, 4, 2, 4, 4, 2, 3, 2, 3, 5, 4, 5, 3, 4, 4...
$ Q11     <dbl> 3, 3, 2, 3, 3, 4, 2, 3, 3, 5, 4, 2, 4, 1, 3, 5, 3, 5, 3, 4, 4...
$ Q12     <dbl> 5, 3, 1, 3, 3, 3, 3, 3, 2, 5, 3, 3, 3, 3, 3, 4, 4, 4, 3, 4, 4...
$ Q13     <dbl> 5, 3, 2, 4, 3, 3, 4, 4, 3, 5, 3, 3, 4, 3, 3, 5, 3, 4, 2, 4, 3...
$ Q14     <dbl> 4, 3, 1, 4, 2, 3, 3, 5, 2, 4, 4, 4, 3, 4, 3, 5, 3, 3, 2, 4, 3...
$ Q15     <dbl> 5, 4, 1, 5, 5, 5, 5, 5, 1, 5, 5, 3, 4, 4, 3, 5, 4, 5, 4, 4, 4...
$ Q16     <dbl> 5, 3, 5, 5, 5, 4, 3, 3, 2, 5, 3, 2, 5, 2, 2, 5, 4, 3, 4, 4, 3...
$ Q17     <dbl> 4, 3, 5, 5, 5, 4, 4, 5, 2, 5, 4, 2, 4, 2, 3, 5, 4, 4, 4, 4, 4...
$ Q18     <dbl> 4, 3, 5, 5, 5, 4, 3, 4, 2, 5, 4, 2, 4, 2, 3, 5, 4, 5, 4, 4, 4...
$ Q19     <dbl> 4, 4, 5, 5, 5, 4, 2, 3, 3, 5, 4, 2, 4, 2, 3, 5, 3, 4, 5, 5, 3...
$ Q20     <dbl> 2, 3, 3, 5, 5, 4, 4, 4, 3, 4, 4, 3, 3, 3, 2, 5, 3, 4, 3, 5, 3...
$ Q21     <dbl> 2, 3, 2, 5, 5, 4, 2, 3, 2, 4, 2, 4, 4, 2, 2, 5, 3, 4, 4, 4, 4...
$ Q22     <dbl> 5, 3, 3, 5, 5, 4, 2, 3, 3, 4, 4, 3, 2, 3, 2, 4, 4, 4, 3, 5, 3...
$ Q23     <dbl> 5, 3, 3, 5, 2, 4, 2, 4, 3, 5, 5, 3, 4, 4, 4, 4, 4, 5, 4, 5, 4...
$ Q24     <dbl> 2, 2, 3, 4, 2, 4, 2, 5, 2, 4, 2, 3, 2, 4, 4, 5, 4, 3, 4, 4, 4...
$ Q25     <dbl> 2, 2, 3, 4, 3, 4, 2, 5, 3, 4, 3, 3, 3, 3, 2, 5, 4, 4, 4, 4, 3...
$ Q26     <dbl> 2, 2, 4, 1, 1, 2, 2, 3, 1, 2, 2, 3, 1, 2, 3, 2, 2, 2, 3, 1, 2...
$ TIME    <dbl> 1200, 1320, 960, 1260, 1140, 1140, 540, 1080, 1380, 660, 1620...
```

Source: [Procedural exclusions](#)

3 Procedural exclusions

Level 1 applies three procedures prior to calculating post-hoc indicators: dropout removal, exclusion for response time below 208 seconds ($104 \text{ items} \times 2 \text{ s/item}$) (Huang et al., 2026), and exclusion for two or more failures on quality control items (CQ1–CQ3) (Curran, 2016). Every exclusion is recorded in `exclusion_log.csv` with its reason and criterion.

Table 1: Procedural exclusions summary

Step	N
Raw data	1546
Dropouts removed	220
Excluded by time ($< 208 \text{ s}$)	0
Excluded by QC (≥ 2 fails)	28
Excluded by missings (> 5 in Q1–Q26)	3
Eligible for Level 2	1295
Remaining NAs in Q1–Q26	90

Source: [Procedural exclusions](#)

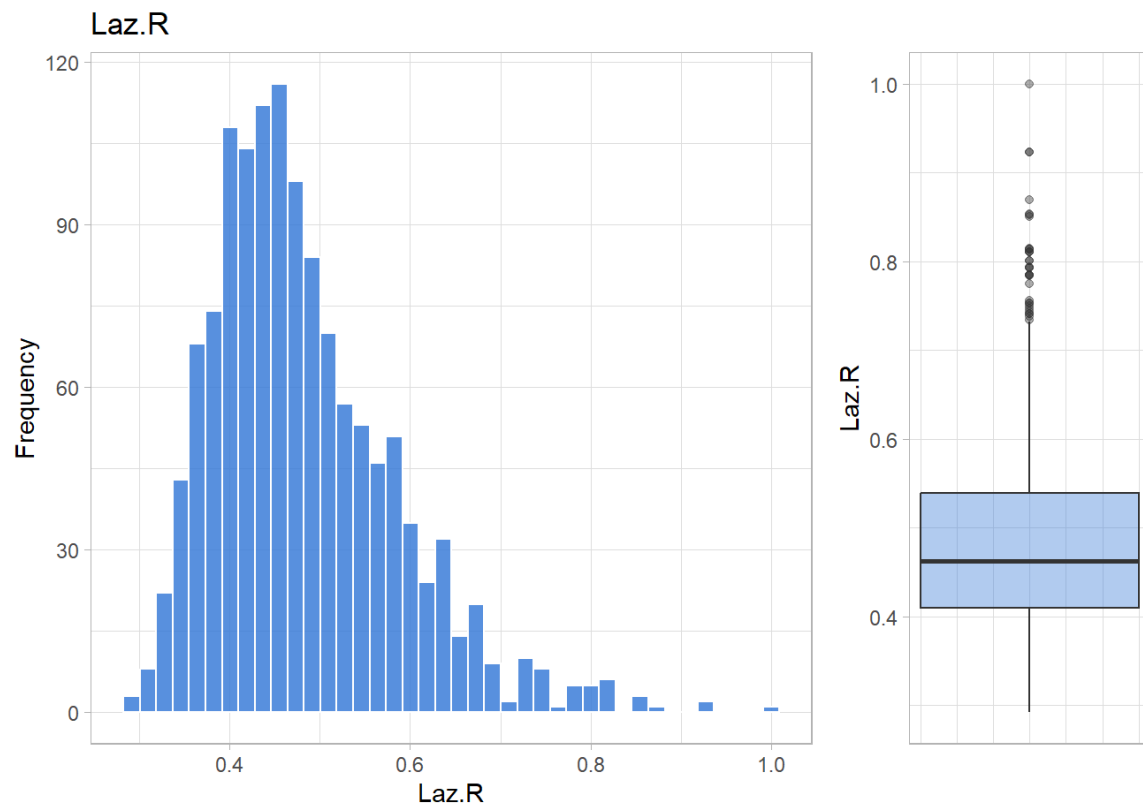
4 Calculation of C/IER indicators

Each indicator captures distinct manifestations of C/IER; their concurrent presence offers stronger evidence than any single indicator alone (Curran, 2016; Hong et al., 2020; Huang et al., 2026).

4.1 Univariate visualization

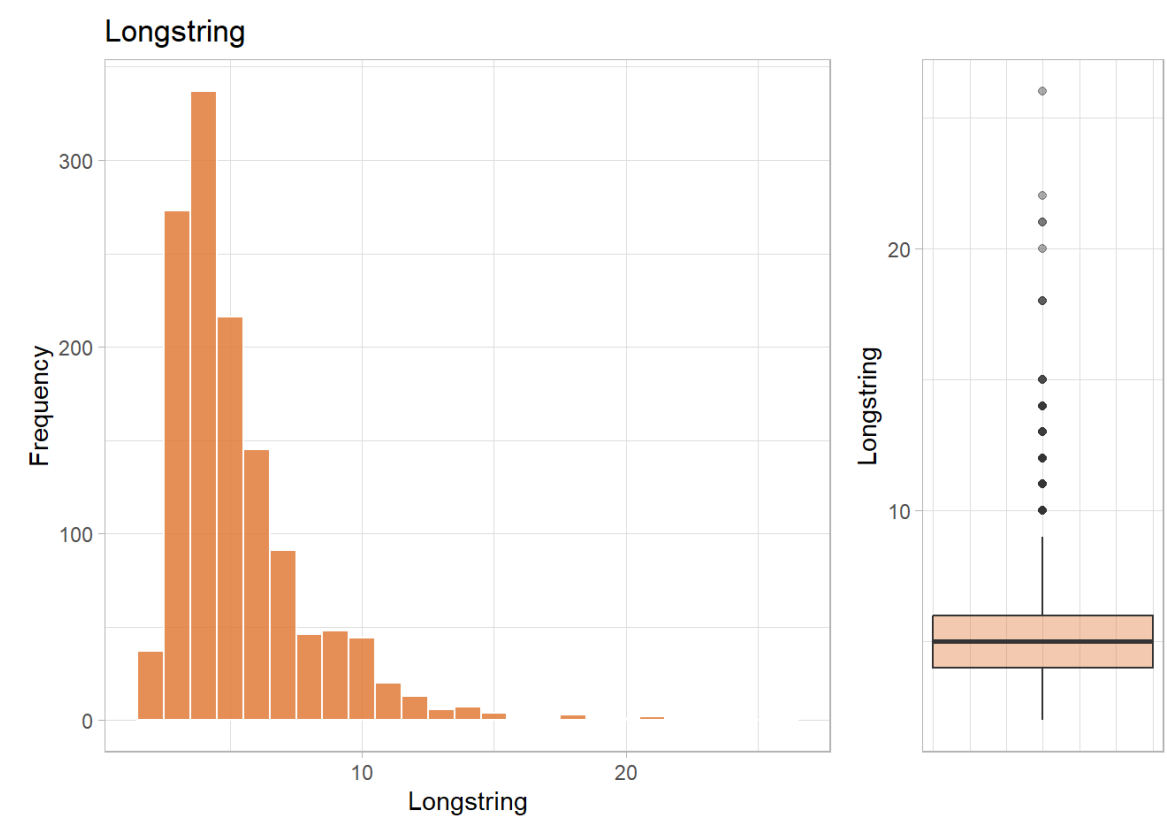
Each respondent receives the values of the four indicators in a consolidated tibble, and the plots below show the univariate distribution of each one.

4.1.1 Laz.R



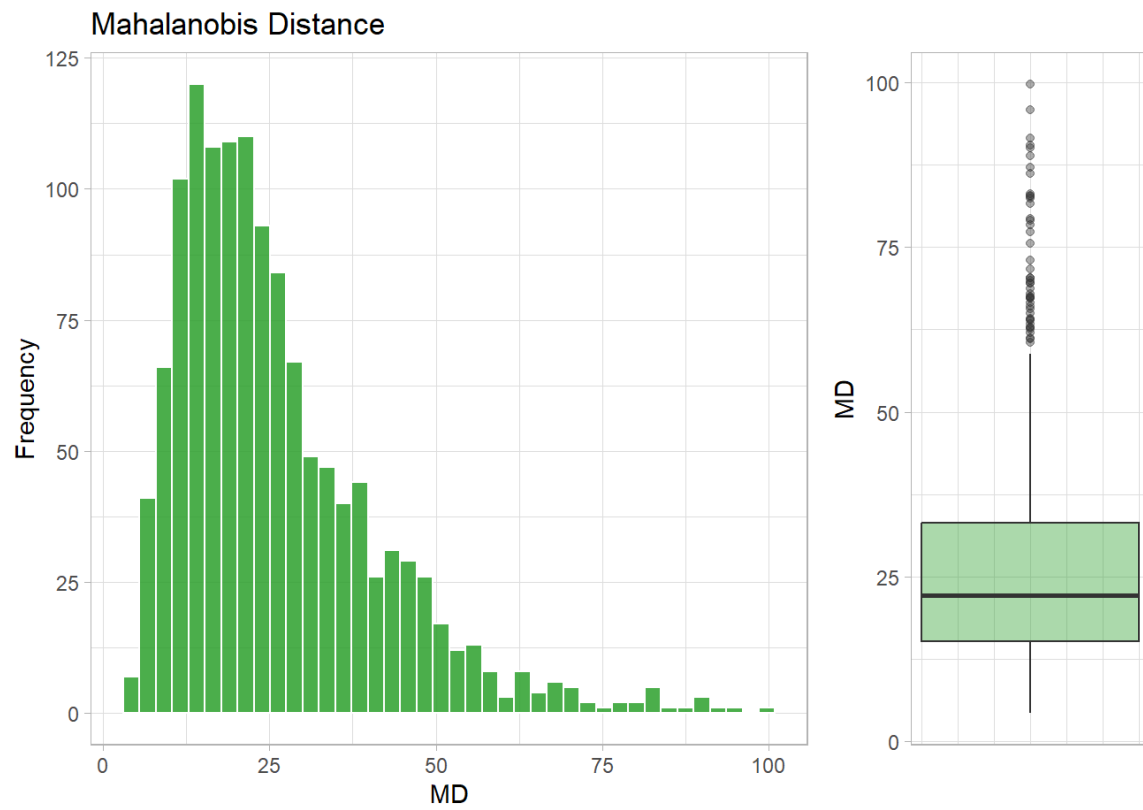
Source: [Post-hoc C/IER Indicators](#)

4.1.2 Longstring



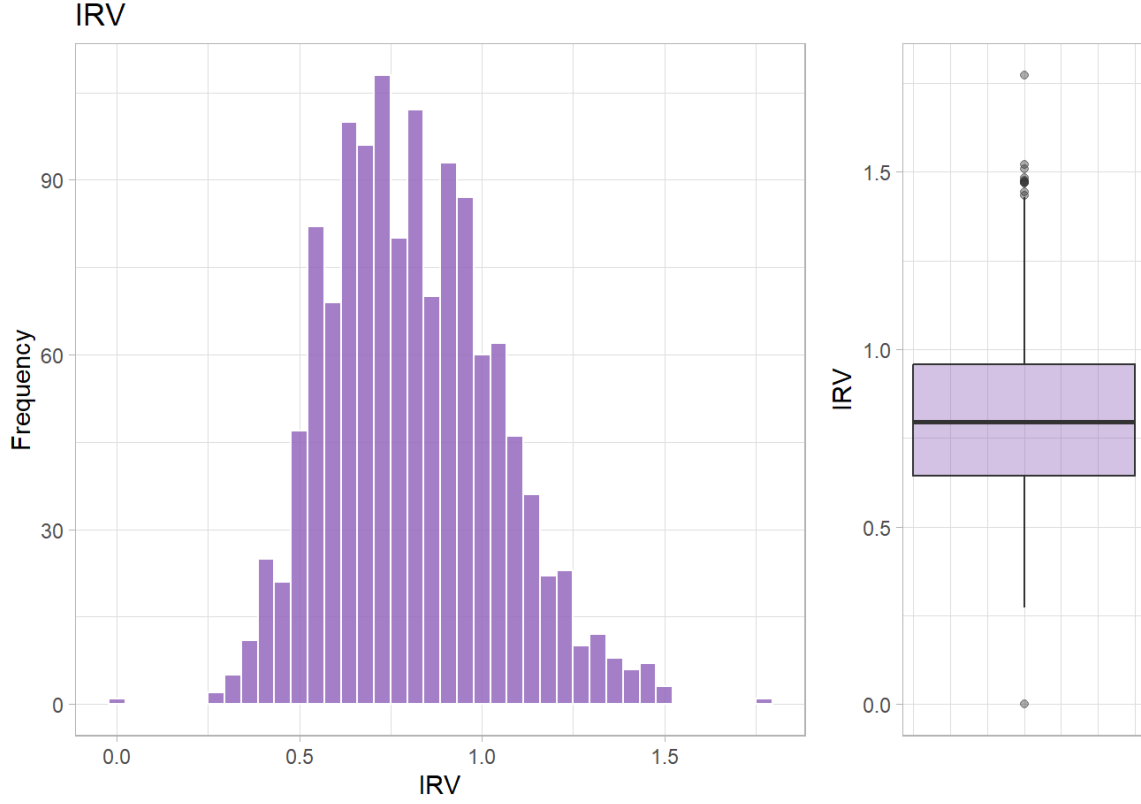
Source: [Post-hoc C/IER Indicators](#)

4.1.3 Mahalanobis Distance



Source: [Post-hoc C/IER Indicators](#)

4.1.4 IRV



Source: [Post-hoc C/IER Indicators](#)

4.2 Cut-off points via Kneedle algorithm

The Kneedle algorithm is applied to the sorted (descending) curve of each indicator. For **Laz.R**, **Longstring**, and **MD**, kneedle with **sign** = -1 identifies where extremely high values separate from the majority (right tail). For **IRV**, **sign** = -1 is used to detect excessively high IRV and **sign** = +1 for excessively low IRV, since both extremes can be considered suspicious.

Table 2: Cut-off points — Kneedle algorithm (Biemann et al., 2025)

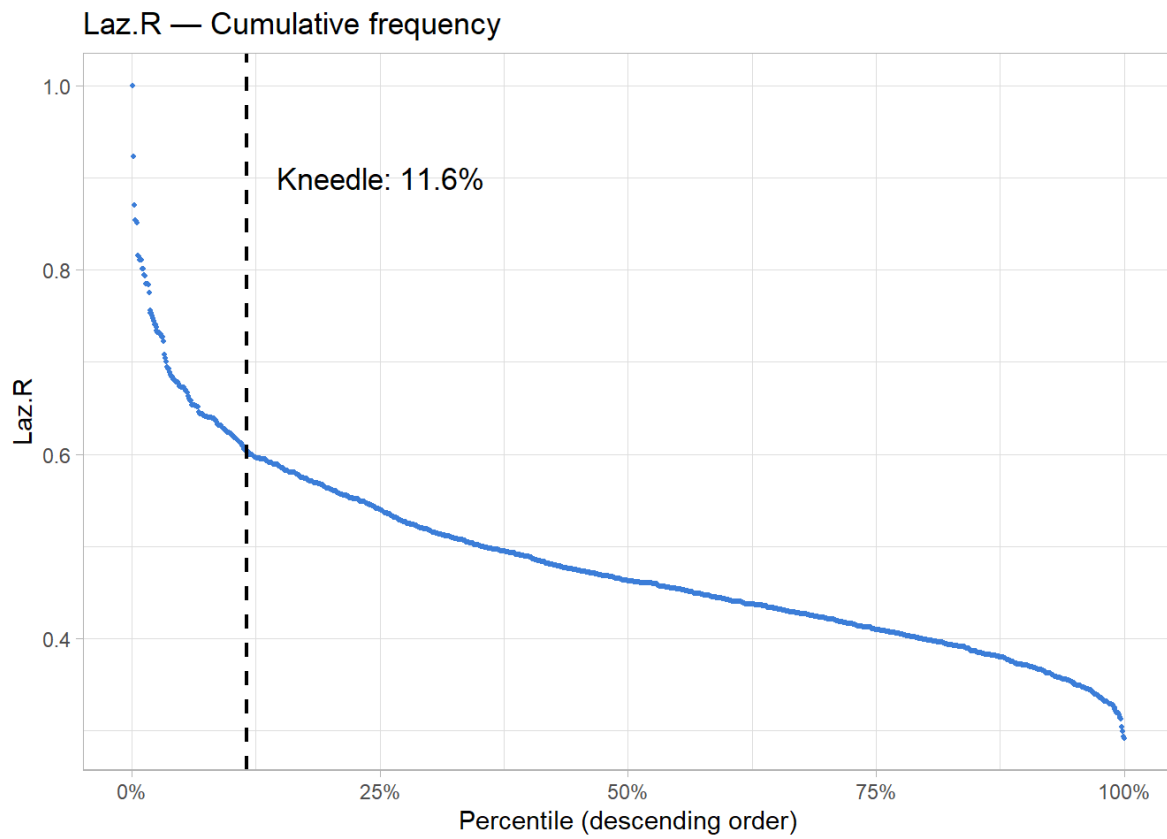
Indicator	Critical value	% flagged
Laz.R	0.60	11.6
Longstring	7.00	15.2
MD	40.12	15.4
IRV (high end)	1.14	8.0

Indicator	Critical value	% flagged
IRV (low end)	0.48	5.0

Source: [Post-hoc C/IER Indicators](#)

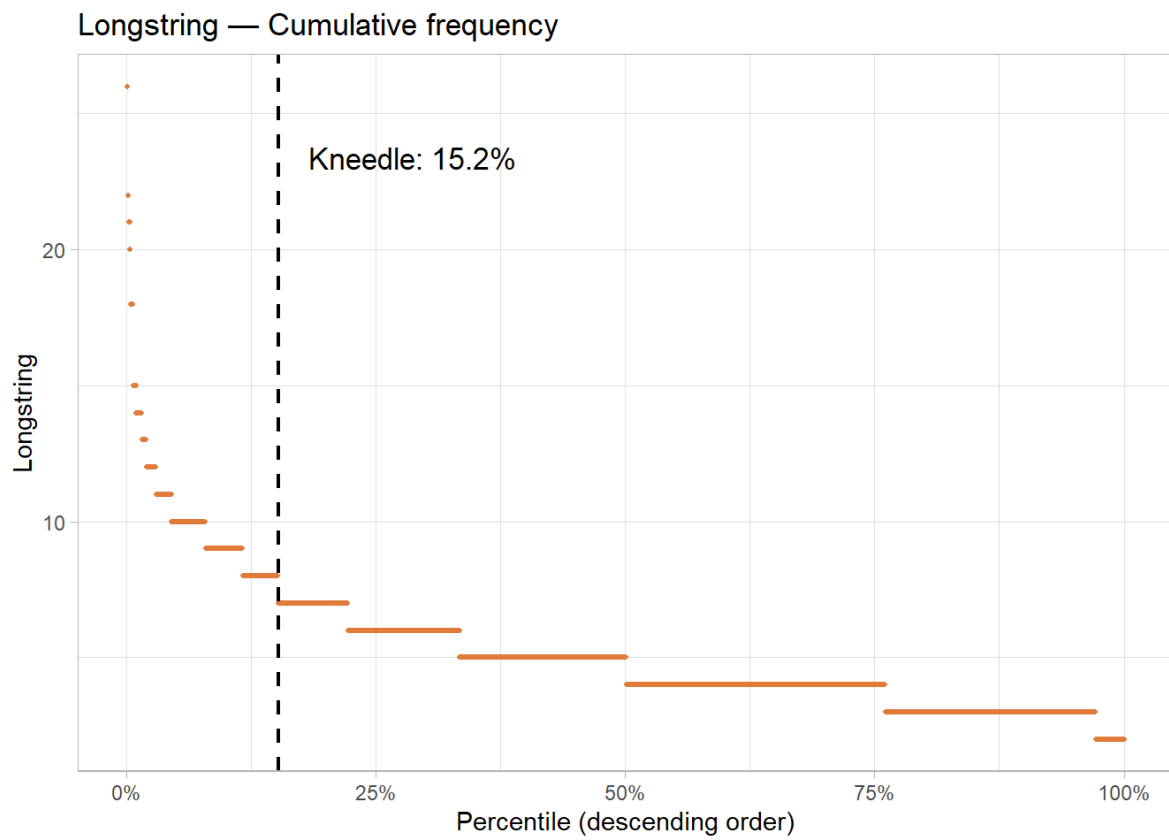
4.3 Plots with cut-off points

4.3.1 Laz.R



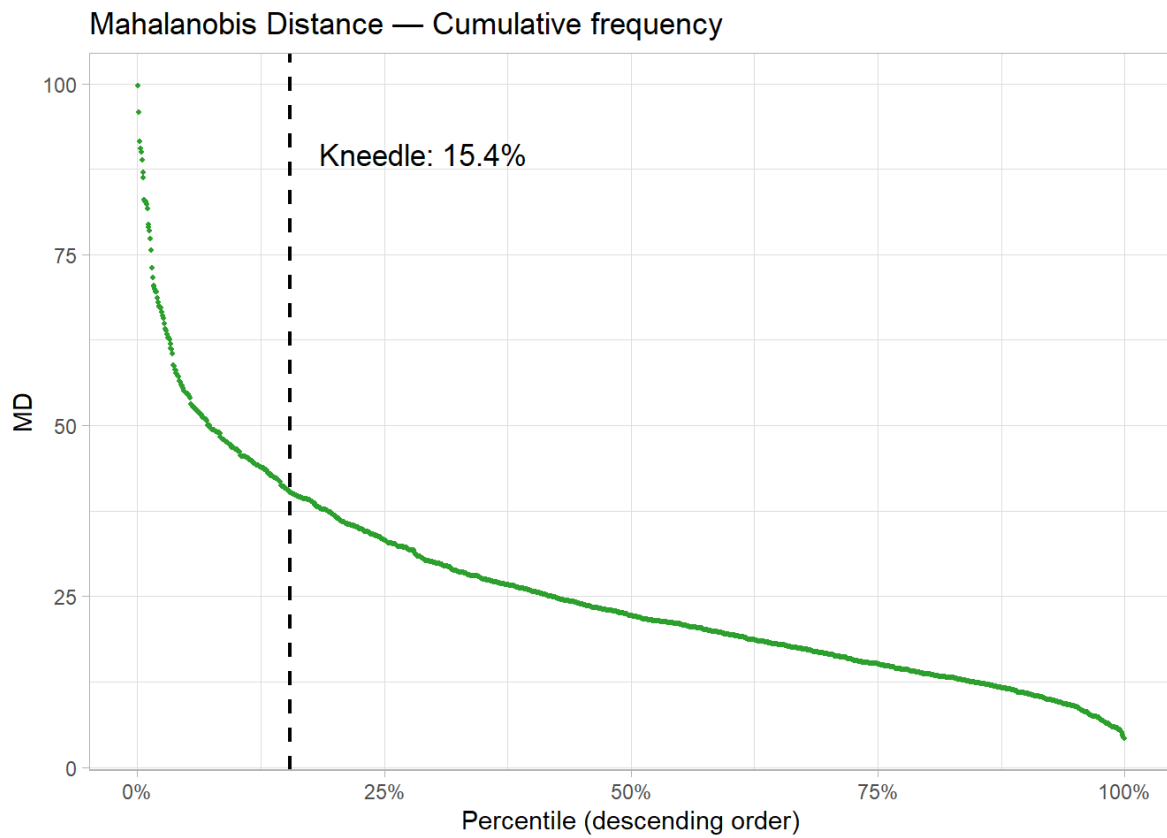
Source: [Post-hoc C/IER Indicators](#)

4.3.2 Longstring



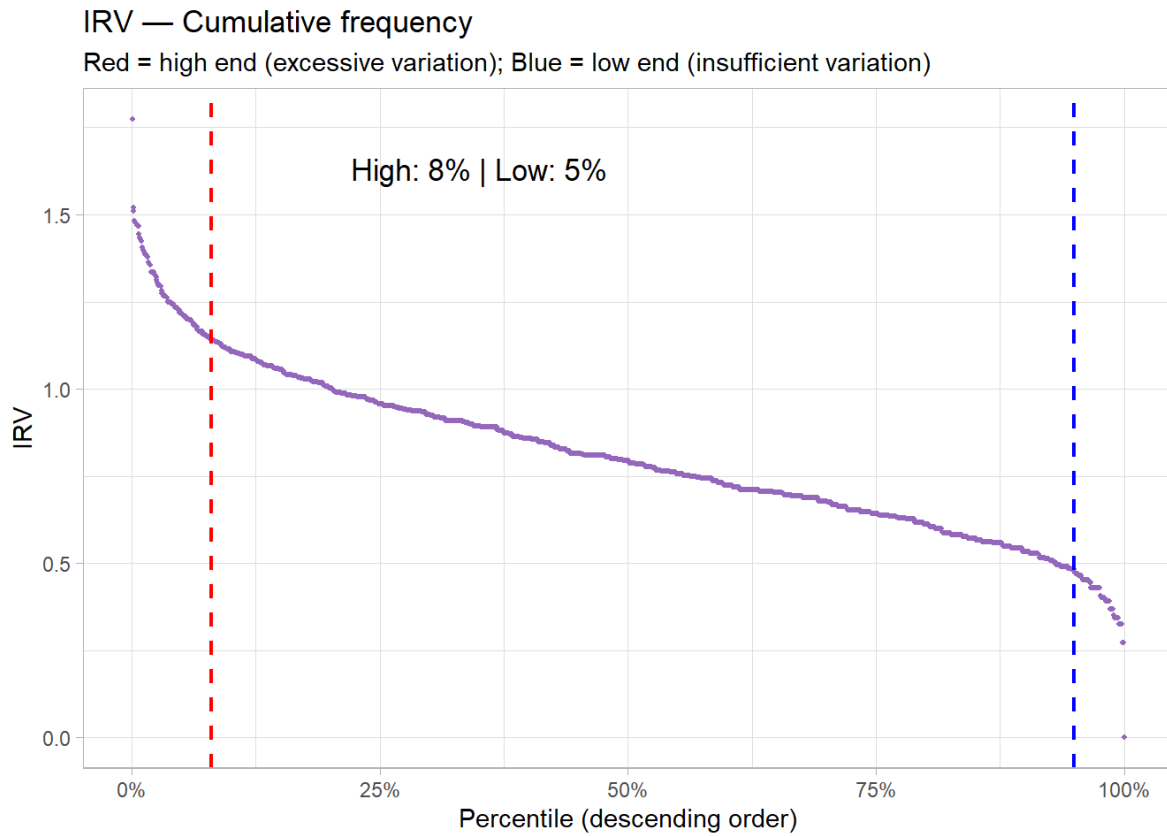
Source: [Post-hoc C/IER Indicators](#)

4.3.3 Mahalanobis Distance



Source: [Post-hoc C/IER Indicators](#)

4.3.4 IRV



Source: [Post-hoc C/IER Indicators](#)

4.4 C/IER Flags

For each indicator, a `_FLAG` variable is created (1 = flagged, 0 = not flagged) based on the Kneedle cut-off points.

Note: Post-hoc indicators are **diagnostic**, they do not trigger automatic exclusion in this phase. The composite flag array will be used for inspection and evaluation through the `whoqol_screened.csv` file.

Table 3: Summary of flagged C/IER respondents

Indicator	N flagged	% flagged
LazR_FLAG	152	11.7
Longstr_FLAG	196	15.1

Indicator	N flagged	% flagged
MD_FLAG	201	15.5
IRV_high_FLAG	108	8.3
IRV_low_FLAG	65	5.0

Source: [Post-hoc C/IER Indicators](#)

5 Multicollinearity

The VIF (Variance Inflation Factor) is calculated for each item in the WHOQOL scale. Values greater than 5 indicate problematic multicollinearity, anticipating convergence and interpretation problems in CFA (Flora et al., 2012; Kline, 2023).

Table 4: Variance Inflation Factor (VIF) per item — VIF > 5 indicates problematic multicollinearity

Item	VIF	Problematic
Q1	2.137	No
Q2	1.979	No
Q3	1.699	No
Q4	1.553	No
Q5	2.252	No
Q6	1.808	No
Q7	1.726	No
Q8	1.323	No
Q9	1.492	No
Q10	2.302	No
Q11	1.627	No
Q12	1.838	No
Q13	1.575	No
Q14	1.973	No
Q15	1.581	No
Q16	1.417	No
Q17	3.580	No
Q18	3.145	No
Q19	2.608	No
Q20	1.888	No
Q21	1.366	No
Q22	1.739	No

Item	VIF	Problematic
Q23	1.434	No
Q24	1.391	No
Q25	1.417	No
Q26	1.566	No

Source: [Multicollinearity](#)

- Biemann, T., Koch-Bayram, I. F., Meier-Barthold, M., & Aguinis, H. (2025). Using Markov Chains to Detect Careless Responding in Survey Research. *Organizational Research Methods*, 28(4), 543–568. <https://doi.org/10.1177/10944281251334778>
- Curran, P. G. (2016). Methods for the detection of carelessly invalid responses in survey data. *Journal of Experimental Social Psychology*, 66, 4–19. <https://doi.org/10.1016/j.jesp.2015.07.006>
- Flora, D. B., LaBrish, C., & Chalmers, R. P. (2012). Old and New Ideas for Data Screening and Assumption Testing for Exploratory and Confirmatory Factor Analysis. *Frontiers in Psychology*, 3(MAR), 55. <https://doi.org/10.3389/fpsyg.2012.00055>
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- Huang, J. L., Wang, Z., Huang, R., Wu, D., & Shi, H. (2026). *Insufficient Effort Responding in Management Research: A Critical Review and Future Directions*. 52(1). <https://doi.org/10.1177/01492063251330268>
- Kline, R. B. (2023). *Principles and Practice of Structural Equation Modeling* (Fifth Edition). The Guilford Press.